DevOps Labs Report

Git & GitHub | Docker Basics | Docker Advanced

1. Git & GitHub - branches, PRs, merge conflict

Repo: github.com/Kubepsychoctl/team-docs-warriors (3 PRs, 2 conflicts resolved)

2. Docker Basics (Week 1)

hello-world, nginx, ubuntu, own Dockerfile, Flask web app; image pushed to Docker Hub as kubernetesson/my-webapp:v1.0

3. Docker Advanced - Lab 1 (Week 4)

react-blog: multi-stage, alpine, non-root; 2.64 GB -> 191 MB (~14x smaller)

4. Docker Advanced - Lab 2 (Week 5)

shop-app: nginx + Flask + Redis on a custom network

Appendix

Earlier local-simulation screenshots of the Git lab

Note: terminal screenshots are rendered from real command output; Docker was run in a sandbox (CA passed as build secret).

1. Git & GitHub lab

Git & GitHub

Pull requests #1-#3 (all merged) - real repo

GitHub PRs for Kubepsychoctl/team-docs-warriors (REST API)

```
$ gh api 'repos/Kubepsychoctl/team-docs-warriors/pulls?state=all' --jq '.[]|"#\(.number) \(.title) | merged=\
(.merged_at!=null) | \(.html_url)"'
#3 Add Carol: team contribution | merged=true | https://github.com/Kubepsychoctl/team-docs-warriors/pull/3
#2 Add Bob: team contribution | merged=true | https://github.com/Kubepsychoctl/team-docs-warriors/pull/2
#1 Add Alice: team contribution | merged=true | https://github.com/Kubepsychoctl/team-docs-warriors/pull/1
```

Git & GitHub

Real merge conflict in PR #2 (README.md)

Real merge conflict in Kubepsychoctl/team-docs-warriors (PR #2)

```
$ git merge main   # on feature/bob, main already contains PR #1
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.

$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

<<<<<<< HEAD
- Bob - CI/CD Engineer

## Technology Stack

- Jenkins
- Linux

## Setup Instructions

1. Install Jenkins
2. Configure pipeline
3. Run first build
=======
- Alice - DevOps Lead

## Technology Stack

- Docker
- Kubernetes

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
>>>>>>> ef2a46d
```

Git & GitHub

git log --oneline --graph --all and final README

Step 7 - git log --oneline --graph --all

```
$ git log --oneline --graph --all
*   c57b978 Merge pull request #3 from Kubepsychoctl/feature/carol
|\
| *   b1b5c9f Resolve merge conflict: combine team contributions
| |\
| |/
|/|
* |   72bad17 Merge pull request #2 from Kubepsychoctl/feature/bob
|\ \
| * \   f54c543 Resolve merge conflict: combine team contributions
| |\ \
| |/ /
|/| |
* | |   ef2a46d Merge pull request #1 from Kubepsychoctl/feature/alice
|\ \ \
| * | | 5414007 Add Alice: team info and setup instructions
|/ / /
| * / fbba07e Add Bob: team info and setup instructions
|/ /
| * bd0b03f Add Carol: team info and setup instructions
|/
* bd55fd3 Add README structure
* 451d473 Initial commit

$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

- Alice - DevOps Lead
- Bob - CI/CD Engineer
- Carol - Cloud Engineer

## Technology Stack

- Docker
- Kubernetes
- Jenkins
- Linux
- Terraform
- AWS

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
4. Install Jenkins
5. Configure pipeline
6. Run first build
7. Install Terraform
8. Run terraform init
9. Run terraform apply
```

2. Docker Basics

Docker Basics

Task 1 - hello-world

```
Task 1 - hello-world

$ docker --version
Docker version 29.9.2, build a82e528

$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
```

Docker Basics

Task 2 - nginx

```
Task 2 - nginx

$ docker run -d -p 8080:80 --name web nginx
Digest: sha256:5ab5b0b01097420873a3f3330f44248e34a3c2a3e8d5ee30658aeae8cb7435da
Status: Downloaded newer image for nginx:latest
866390c8209042a185258008b2a14f01a3386dacb2f69988c6f0ba82f28c0187

$ curl -s http://localhost:8080 | head -n 12
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>

$ docker logs web | tail -n 6
2026/11/03 16:04:23 [notice] 1#1: start worker processes
2026/11/03 16:04:23 [notice] 1#1: start worker process 20
2026/11/03 16:04:23 [notice] 1#1: start worker process 21
2026/11/03 16:04:23 [notice] 1#1: start worker process 22
2026/11/03 16:04:23 [notice] 1#1: start worker process 23
172.17.0.1 - - [03/Nov/2026:16:04:26 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.5.0" "-"

$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                  NAMES
866390c82090   nginx     "/docker-entrypoint.…"   4 seconds ago   Up 3 seconds   0.0.0.0:8080->80/tcp   web

$ docker stop web && docker rm web
web
web
```

Docker Basics

Task 3 - ubuntu container

```
Task 3 - ubuntu container

$ docker run -it --name my-ubuntu ubuntu bash   # commands fed non-interactively
[os-release]
PRETTY_NAME="Ubuntu 26.04.1 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"

$ ls /
bin
boot
dev
etc
home
lib
lib64
media
mnt
opt
proc
root
run
sbin
srv
sys
tmp
usr
var
Hello

$ docker rm my-ubuntu
my-ubuntu
```

Docker Basics

Task 4 - own Dockerfile (hello-docker)

Task 4 - own Dockerfile

```
$ cd hello-docker && docker build -t hello-docker .
#7 DONE 0.0s

#8 exporting to image
#8 exporting layers 0.1s done
#8 exporting manifest sha256:e6202534159b2c3490837e0ddf8ddc0b1e58cad884ffcee6c5c0af0091baba42 done
#8 exporting config sha256:33918b8c39c632e92a773d62bd9972a076da852ebfdc741e8d076b853cff5a7e done
#8 exporting attestation manifest sha256:be1f8fccd53d4d012bc90bd5c4c630c104b49ed637a2e644e5570b89134c5a11 0.0s done
#8 exporting manifest list sha256:415be81095c9c0ed64544e2c64cda080c057b7e412901f51958af0a651a84d3f done
#8 naming to docker.io/library/hello-docker:latest done
#8 unpacking to docker.io/library/hello-docker:latest
#8 unpacking to docker.io/library/hello-docker:latest 1.5s done
#8 DONE 1.6s

$ docker run --rm hello-docker
==================================================
Привет из Docker!
Моё имя: Kubepsychoctl
Этот код работает в контейнере!
==================================================
```

Docker Basics

Lab - Flask my-webapp build and run

Lab - Flask my-webapp

```
$ docker build -t my-webapp .
#3 DONE 0.0s
#4 DONE 0.0s
#5 DONE 0.0s
#8 6.567 Successfully installed Flask-3.0.0 Jinja2-3.1.6 MarkupSafe-3.0.4 Werkzeug-3.1.9 blinker-1.9.0 click-8.5.0
itsdangerous-2.2.0
#8 DONE 6.9s
#9 DONE 0.0s
#10 naming to docker.io/library/my-webapp:latest done
#10 DONE 1.2s

$ docker run -d -p 5000:5000 --name webapp my-webapp
8d35399b9f3e8fa04ac595edac323d187aed36182f5f9d5a04d7e877d08393fb

$ curl -s http://localhost:5000

    <html>
    <body style="font-family: Arial; text-align: center; padding: 50px;">
        <h1 style="color: #0E4C92;">🐳 Привет из Docker!</h1>
        <p style="font-size: 20px;">
            <strong>Меня зовут:</strong> Kubepsychoctl
        </p>
        <p style="color: gray;">
            Время на сервере: 2026-11-03 16:05:21
        </p>
        <hr>
        <p style="font-size: 14px;">
            Этот сервер работает в Docker контейнере ✅
        </p>
    </body>
    </html>

$ docker ps
CONTAINER ID   IMAGE       COMMAND           CREATED         STATUS         PORTS                    NAMES
8d35399b9f3e   my-webapp   "python app.py"   4 seconds ago   Up 4 seconds   0.0.0.0:5000->5000/tcp   webapp

$ docker images
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-docker:latest   415be81095c9        187MB         45.6MB
hello-world:latest    c567b231f656       25.9kB         9.49kB   U
my-webapp:latest      d1d77635a9a5        211MB         51.3MB   U
nginx:latest          5ab5b0b01097        243MB         66.8MB
ubuntu:latest         6a06111c3b0a        163MB         45.9MB
```

Docker Basics

my-webapp in the browser

🐳 Привет из Docker!

Меня зовут: Kubepsychoctl

Время на сервере: 2026-11-03 16:05:34

Этот сервер работает в Docker контейнере ✅

3. Docker Advanced - Lab 1

Docker Advanced - Lab 1 (react-blog)

Image size: naive 2.64 GB vs optimized 191 MB

```
Lab 1 - image size: naive vs optimized (multi-stage)

$ docker images | grep react-blog
IMAGE                     ID             DISK USAGE   CONTENT SIZE   EXTRA
react-blog-naive:latest   eda256c34f8b       2.64GB          561MB
react-blog:latest         0a3730d56616        191MB         46.8MB   U

naive:     node:18, COPY . ., no .dockerignore, single stage = 2.64 GB
optimized: multi-stage + alpine + non-root + .dockerignore = 191 MB (~14x smaller, < 200 MB)
```

Docker Advanced - Lab 1 (react-blog)

Run, non-root user, health check, layers

```
Lab 1 - run, non-root user, health check

$ docker images | grep react-blog
IMAGE                     ID             DISK USAGE   CONTENT SIZE   EXTRA
react-blog-naive:latest   eda256c34f8b       2.64GB          561MB
react-blog:latest         0a3730d56616        191MB         46.8MB

$ docker run -d -p 3000:3000 --name blog react-blog
135f873347e1114e315473d894bf451b19f9133ff60b02552cc6061718aab9bc

$ docker exec blog whoami   # non-root check
appuser
uid=1001(appuser) gid=1001(appgroup) groups=1001(appgroup)

$ curl -s localhost:3000/api/health
{"status":"ok","message":"Server is running"}

$ curl -s localhost:3000 | head -c 250
<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="icon" href="/favicon.ico"/><meta name="viewport" cont
ent="width=device-width,initial-scale=1"/><meta name="theme-color" content="#000000"/><meta name="description" content="Web
si

$ docker logs blog
Server running on port 3000

$ docker history react-blog --format "{{.Size}}\t{{.CreatedBy}}" | head -n 8
0B      CMD ["node" "server.js"]
0B      EXPOSE [3000/tcp]
0B      USER appuser
1.39MB  COPY /app/client/build ./build # buildkit
12.3kB  COPY server/server.js . # buildkit
6.99MB  RUN /bin/sh -c NODE_EXTRA_CA_CERTS=/tmp/ca.c…
45.1kB  COPY server/package*.json ./ # buildkit
8.19kB  WORKDIR /app
```

Docker Advanced - Lab 1 (react-blog)

react-blog in the browser

📝 Docker Blog

Изучаем Docker

Multi-stage builds

Best Practices

Оптимизация образов

4. Docker Advanced - Lab 2

Docker Advanced - Lab 2 (shop-app)

./start.sh - network and 3 containers

Lab 2 - ./start.sh (network + 3 containers)

```
$ ./start.sh
a689950e4b8f621e1007e8d4c4b0783090bb5f03bb34e78dca9b45785af5f9f0
Unable to find image 'redis:7-alpine' locally
7-alpine: Pulling from library/redis
e25d07126ba7: Pulling fs layer
cef7ad5a05c6: Pulling fs layer
759efa4719c9: Pulling fs layer
a239ec2b8fe1: Pulling fs layer
ae75cb664c8f: Pulling fs layer
1ea868e596ec: Pulling fs layer
4f4fb700ef54: Pulling fs layer
4f4fb700ef54: Already exists
b5e4a0ae9c7c: Download complete
4d584b19579f: Download complete
cef7ad5a05c6: Download complete
759efa4719c9: Download complete
e25d07126ba7: Download complete
a239ec2b8fe1: Download complete
ae75cb664c8f: Download complete
759efa4719c9: Pull complete
a239ec2b8fe1: Pull complete
cef7ad5a05c6: Pull complete
1ea868e596ec: Download complete
ae75cb664c8f: Pull complete
1ea868e596ec: Pull complete
4f4fb700ef54: Pull complete
e25d07126ba7: Pull complete
Digest: sha256:8da19c7399dca900c6822a1cf30f9d2285585e2c4b457ff4095b0f4500054d3f
Status: Downloaded newer image for redis:7-alpine
08073f7b7955548ca2e0389eb25d16eee0a13efbbdb19c4a03bdd193d7de8225
95371a2c83d9a27cd686396c1050fcd1fc2a1842f3d21c0e6f021e1ab6f8f879
96a914695f25cf27b49dc447f0df87db544c08cfb61313c41198ea54a0fcbfba
Shop запущен!
Frontend: http://localhost:8080
Backend API: http://localhost:5000/products
exit=0
```

Docker Advanced - Lab 2 (shop-app)

Network, DNS by name, API, Redis

```
Lab 2 - network, DNS by name, API, Redis

$ docker ps
NAMES      IMAGE            PORTS                    STATUS
frontend   shop-frontend    0.0.0.0:8080->80/tcp     Up 5 seconds
backend    shop-backend     0.0.0.0:5000->5000/tcp   Up 10 seconds
redis      redis:7-alpine   6379/tcp                 Up 16 seconds

$ docker network ls | grep shop
NETWORK ID     NAME           DRIVER    SCOPE
a689950e4b8f   shop-network   bridge    local

$ docker network inspect shop-network --format "{{range .Containers}}{{.Name}} {{.IPv4Address}}{{println}}{{end}}"
redis 172.18.0.2/16
backend 172.18.0.3/16
frontend 172.18.0.4/16

$ docker exec backend python -c "import socket;print(socket.gethostbyname(\"redis\"))"   # DNS by name
redis -> 172.18.0.2

$ curl -s localhost:5000/products
[{"id":1,"name":"Docker T-Shirt","price":25},{"id":2,"name":"Kubernetes Mug","price":15},{"id":3,"name":"DevOps Book","pric
e":40}]

$ curl -s -X POST localhost:5000/cart/add/1 ; ...add/3
{"success":true}
{"success":true}

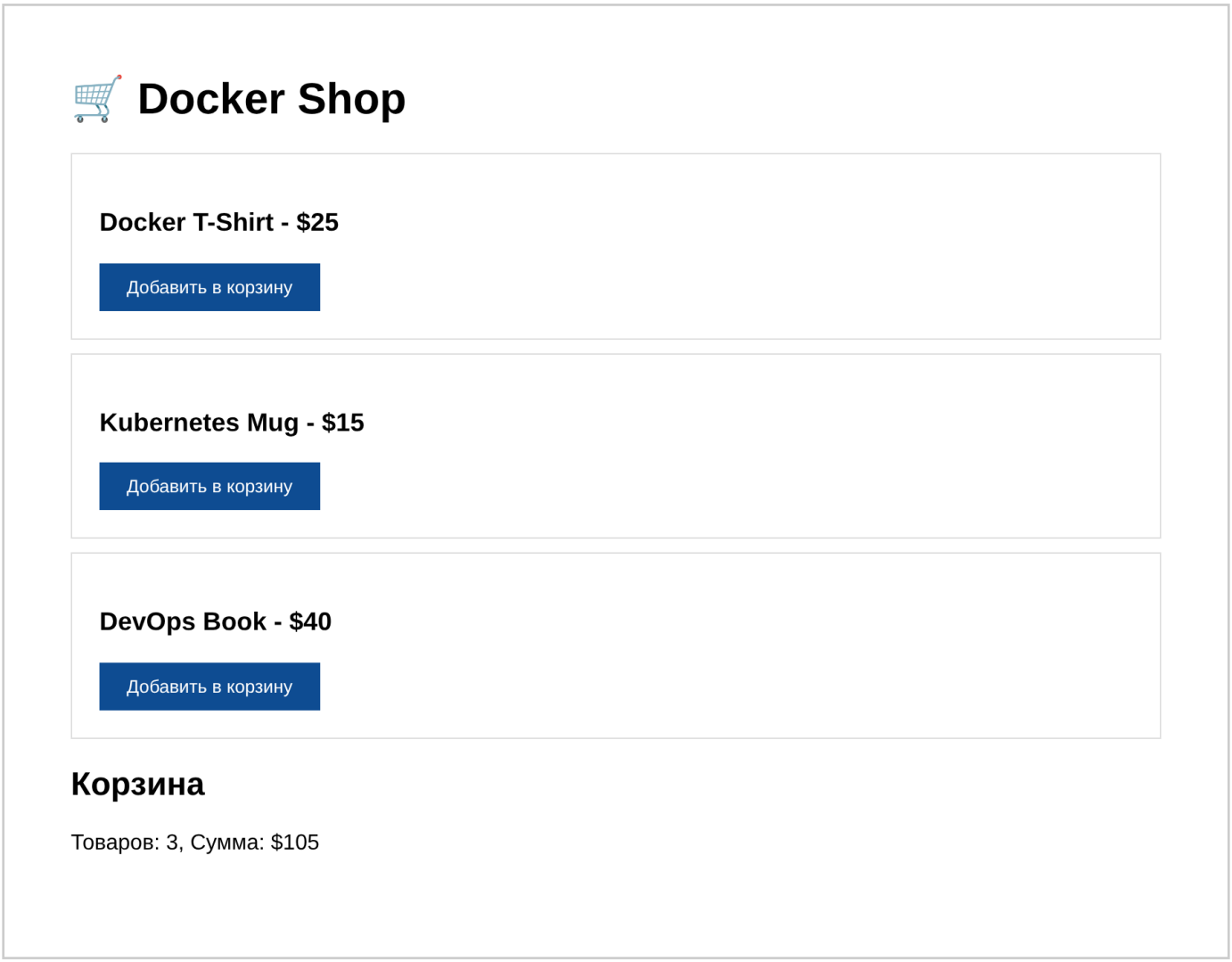
$ curl -s localhost:5000/cart
{"count":2,"items":[{"id":3,"name":"DevOps Book","price":40},{"id":1,"name":"Docker T-Shirt","price":25}],"total":65}

$ docker exec redis redis-cli LLEN cart
2

$ curl -s localhost:8080 | head -n 4
<!DOCTYPE html>
<html>
<head>
  <title>Docker Shop</title>
```

Docker Advanced - Lab 2 (shop-app)

Docker Shop with cart in the browser



Appendix - Git lab local simulation

Appendix

Step 1 (local simulation)

```
Step 1 - Create repository with README structure (Owner: Alice)

@team-docs$ git init --bare -b main /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origi
n.git
Initialized empty Git repository in /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origi
n.git/
@team-docs$ git clone /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git team-docs
-devops-team
Cloning into 'team-docs-devops-team'...
warning: You appear to have cloned an empty repository.
done.
alice@team-docs$ git config user.name "Alice" && git config user.email "alice@example.com"
alice@team-docs$ git add README.md && git commit -m 'Initial commit' && git push -u origin main
[main (root-commit) 968a7a8] Initial commit
 1 file changed, 17 insertions(+)
 create mode 100644 README.md
branch 'main' set up to track 'origin/main'.
fatal: expected 'acknowledgments', received 'packfile'
warning: push negotiation failed; proceeding anyway with push
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
 * [new branch]      main -> main
alice@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

List of team members here...

## Technology Stack

Technologies here...

## Setup Instructions

Steps to run the project here...
```

Appendix

Step 2 (local simulation)

```
Step 2 - Clone and create personal branches

@team-docs$ git clone /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git team-docs
-devops-team
Cloning into 'team-docs-devops-team'...
done.
bob@team-docs$ git config user.name "Bob" && git config user.email "bob@example.com"
@team-docs$ git clone /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git team-docs
-devops-team
Cloning into 'team-docs-devops-team'...
done.
carol@team-docs$ git config user.name "Carol" && git config user.email "carol@example.com"
alice@team-docs$ git checkout -b feature/alice
Switched to a new branch 'feature/alice'
alice@team-docs$ git branch
* feature/alice
  main
bob@team-docs$ git checkout -b feature/bob
Switched to a new branch 'feature/bob'
bob@team-docs$ git branch
* feature/bob
  main
carol@team-docs$ git checkout -b feature/carol
Switched to a new branch 'feature/carol'
carol@team-docs$ git branch
* feature/carol
  main
```

Appendix

Step 3 (local simulation)

```
Step 3 - Each member edits the SAME README.md

alice@team-docs$ git diff
diff --git a/README.md b/README.md
index 9a652b0..404f226 100644
--- a/README.md
+++ b/README.md
@@ -6,12 +6,15 @@ General description here...

 ## Team

-List of team members here...
+- Alice - DevOps Lead

 ## Technology Stack

-Technologies here...
+- Docker
+- Kubernetes

 ## Setup Instructions

-Steps to run the project here...
+1. Install Docker
+2. Run docker compose up
+3. Open localhost:8080
bob@team-docs$ git diff
diff --git a/README.md b/README.md
index 9a652b0..bc9ee1d 100644
--- a/README.md
+++ b/README.md
@@ -6,12 +6,15 @@ General description here...

 ## Team

-List of team members here...
+- Bob - CI/CD Engineer

 ## Technology Stack

-Technologies here...
+- Jenkins
+- Linux

 ## Setup Instructions

-Steps to run the project here...
+1. Install Jenkins
+2. Configure pipeline
+3. Run first build
carol@team-docs$ git diff
diff --git a/README.md b/README.md
index 9a652b0..e83e40e 100644
--- a/README.md
+++ b/README.md
@@ -6,12 +6,15 @@ General description here...

 ## Team

-List of team members here...
+- Carol - Cloud Engineer

 ## Technology Stack

-Technologies here...
+- Terraform
+- AWS

 ## Setup Instructions

-Steps to run the project here...
+1. Install Terraform
+2. Run terraform init
+3. Run terraform apply
```

Appendix

Step 4 (local simulation)

Step 4 - Commit and push

```
alice@team-docs$ git status
On branch feature/alice
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
alice@team-docs$ git add README.md
alice@team-docs$ git commit -m "Add Alice: team info and setup instructions"
[feature/alice 72eb7b6] Add Alice: team info and setup instructions
 1 file changed, 6 insertions(+), 3 deletions(-)
alice@team-docs$ git push origin feature/alice
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
 * [new branch]      feature/alice -> feature/alice
bob@team-docs$ git status
On branch feature/bob
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
bob@team-docs$ git add README.md
bob@team-docs$ git commit -m "Add Bob: team info and setup instructions"
[feature/bob 919f4a7] Add Bob: team info and setup instructions
 1 file changed, 6 insertions(+), 3 deletions(-)
bob@team-docs$ git push origin feature/bob
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
 * [new branch]      feature/bob -> feature/bob
carol@team-docs$ git status
On branch feature/carol
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
carol@team-docs$ git add README.md
carol@team-docs$ git commit -m "Add Carol: team info and setup instructions"
[feature/carol b91eb59] Add Carol: team info and setup instructions
 1 file changed, 6 insertions(+), 3 deletions(-)
carol@team-docs$ git push origin feature/carol
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
 * [new branch]      feature/carol -> feature/carol
```

Appendix

Step 5 (local simulation)

```
Step 5 - Pull Requests (simulated; no GitHub UI available)

On GitHub each branch gets a PR titled 'Add [name]: team contribution'. Here PR = merge of the branch into main with a 'Merg
e pull request #N' commit.
alice@team-docs$ git branch -r
  origin/feature/alice
  origin/main
```

Appendix

Step 6 - Bob's conflict (local simulation) (part 1/2)

```
Step 6 - Merge PR #1, then conflict on PR #2

alice@team-docs$ git fetch origin
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * [new branch]      feature/bob   -> origin/feature/bob
 * [new branch]      feature/carol -> origin/feature/carol
alice@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
alice@team-docs$ git checkout main && git pull origin main
Your branch is up to date with 'origin/main'.
Already up to date.
Already on 'main'
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
alice@team-docs$ git merge --no-ff feature/alice -m "Merge pull request #1 - feature/alice" && git push origin main
Merge made by the 'ort' strategy.
 README.md | 9 ++++++---
 1 file changed, 6 insertions(+), 3 deletions(-)
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   968a7a8..feb6df7  main -> main
alice@team-docs$ git merge --no-ff origin/feature/bob -m 'Merge pull request #2 - feature/bob'
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
GitHub would show: 'This branch has conflicts that must be resolved' -> abort and let Bob fix it on his branch.
alice@team-docs$ git merge --abort
bob@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
bob@team-docs$ git pull origin main
Updating 968a7a8..feb6df7
Fast-forward
 README.md | 9 ++++++---
 1 file changed, 6 insertions(+), 3 deletions(-)
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
   968a7a8..feb6df7  main       -> origin/main
bob@team-docs$ git checkout feature/bob
Switched to branch 'feature/bob'
bob@team-docs$ git merge main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
bob@team-docs$ git status
On branch feature/bob
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
bob@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

<<<<<<< HEAD
- Bob - CI/CD Engineer

## Technology Stack

- Jenkins
- Linux

## Setup Instructions

1. Install Jenkins
2. Configure pipeline
3. Run first build
=======
```

Appendix

Step 6 - Bob's conflict (local simulation) (part 2/2)

```
- Alice - DevOps Lead

## Technology Stack

- Docker
- Kubernetes

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
>>>>>>> main
bob@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

- Alice - DevOps Lead
- Bob - CI/CD Engineer

## Technology Stack

- Docker
- Kubernetes
- Jenkins
- Linux

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
4. Install Jenkins
5. Configure pipeline
6. Run first build
bob@team-docs$ grep -nE '^(<<<<<<<|=======|>>>>>>>)' README.md || echo 'no conflict markers left'
no conflict markers left
bob@team-docs$ git add README.md
bob@team-docs$ git commit -m "Resolve merge conflict: combine team contributions"
[feature/bob 54e32db] Resolve merge conflict: combine team contributions
bob@team-docs$ git push origin feature/bob
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   919f4a7..54e32db  feature/bob -> feature/bob
alice@team-docs$ git checkout main && git pull origin main && git fetch origin
Your branch is up to date with 'origin/main'.
Already up to date.
Already on 'main'
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
   919f4a7..54e32db  feature/bob -> origin/feature/bob
alice@team-docs$ git merge --no-ff origin/feature/bob -m 'Merge pull request #2 - feature/bob' && git push origin main
Merge made by the 'ort' strategy.
 README.md | 6 ++++++
 1 file changed, 6 insertions(+)
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   feb6df7..df7ae91  main -> main
```

Appendix

Step 6 - Carol's conflict (local simulation) (part 1/2)

```
Step 6b - Same conflict for PR #3 (Carol)

carol@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
carol@team-docs$ git pull origin main
Updating 968a7a8..df7ae91
Fast-forward
 README.md | 15 ++++++++++---
 1 file changed, 12 insertions(+), 3 deletions(-)
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
   968a7a8..df7ae91  main       -> origin/main
carol@team-docs$ git checkout feature/carol
Switched to branch 'feature/carol'
carol@team-docs$ git merge main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
carol@team-docs$ git status
On branch feature/carol
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
carol@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

- Alice - DevOps Lead
- Bob - CI/CD Engineer
- Carol - Cloud Engineer

## Technology Stack

- Docker
- Kubernetes
- Jenkins
- Linux
- Terraform
- AWS

## Setup Instructions

1. Install Docker
2. Run docker compose up
3. Open localhost:8080
4. Install Jenkins
5. Configure pipeline
6. Run first build
7. Install Terraform
8. Run terraform init
9. Run terraform apply
carol@team-docs$ git add README.md
carol@team-docs$ git commit -m "Resolve merge conflict: combine team contributions"
[feature/carol 585cf10] Resolve merge conflict: combine team contributions
carol@team-docs$ git push origin feature/carol
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   b91eb59..585cf10  feature/carol -> feature/carol
alice@team-docs$ git checkout main && git pull origin main && git fetch origin
Your branch is up to date with 'origin/main'.
Already up to date.
Already on 'main'
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
   b91eb59..585cf10  feature/carol -> origin/feature/carol
alice@team-docs$ git merge --no-ff origin/feature/carol -m 'Merge pull request #3 - feature/carol' && git push origin main
```

Appendix

Step 6 - Carol's conflict (local simulation) (part 2/2)

```
Merge made by the 'ort' strategy.
 README.md | 6 ++++++
 1 file changed, 6 insertions(+)
To /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin.git
   df7ae91..8b66926  main -> main
```

Appendix

Step 7 - final history (local simulation) (part 1/2)

```
Step 7 - Sync and final history

alice@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Already on 'main'
alice@team-docs$ git pull origin main
Already up to date.
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
bob@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
bob@team-docs$ git pull origin main
Updating feb6df7..8b66926
Fast-forward
 README.md | 12 ++++++++++++
 1 file changed, 12 insertions(+)
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
   feb6df7..8b66926  main       -> origin/main
carol@team-docs$ git checkout main
Your branch is up to date with 'origin/main'.
Switched to branch 'main'
carol@team-docs$ git pull origin main
Updating df7ae91..8b66926
Fast-forward
 README.md | 6 ++++++
 1 file changed, 6 insertions(+)
From /tmp/claude-0/-home-user/37796884-530a-5e00-bfa6-e6aab39bef4a/scratchpad/lab/work/origin
 * branch            main       -> FETCH_HEAD
   df7ae91..8b66926  main       -> origin/main
alice@team-docs$ git log --oneline --graph --all
*   8b66926 Merge pull request #3 - feature/carol
|\
| *   585cf10 Resolve merge conflict: combine team contributions
| |\
| |/
|/|
* |   df7ae91 Merge pull request #2 - feature/bob
|\ \
| * \   54e32db Resolve merge conflict: combine team contributions
| |\ \
| |/ /
|/| |
* | |   feb6df7 Merge pull request #1 - feature/alice
|\ \ \
| * | | 72eb7b6 Add Alice: team info and setup instructions
|/ / /
| * / 919f4a7 Add Bob: team info and setup instructions
|/ /
| * b91eb59 Add Carol: team info and setup instructions
|/
* 968a7a8 Initial commit
alice@team-docs$ cat README.md
# Our DevOps Project

## Project Description

General description here...

## Team

- Alice - DevOps Lead
- Bob - CI/CD Engineer
- Carol - Cloud Engineer

## Technology Stack

- Docker
- Kubernetes
- Jenkins
- Linux
- Terraform
- AWS

## Setup Instructions

1. Install Docker
```

Appendix

Step 7 - final history (local simulation) (part 2/2)

```
2. Run docker compose up
3. Open localhost:8080
4. Install Jenkins
5. Configure pipeline
6. Run first build
7. Install Terraform
8. Run terraform init
9. Run terraform apply
```